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### Fluid connector system

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#### Abstract

Fluid connector systems that can include first and second connectors that are couplable together to form a fluid pathway through the fluid connector system when the first and second connectors are coupled together, and can resist fluid flow through each of the first and second connectors when the first and second connectors are separated from each other, where the first and second connectors can include a valve within a channel and configured resist fluid flow through the respective first or second connector in an closed position and to reduce the resistance to fluid flow through the first or second connector in an open position, and the first and second connectors including one or more arm configured to engage each other to form a snap fitting feature that can resist separation therebetween.

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## **Background/Summary**

CROSS REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of priority under 35 U.S.C. § 119 to U.S. Provisional Patent Application Ser. No. 63/397,139, entitled “FLUID CONNECTOR SYSTEM,” filed on Aug. 11, 2022, the entire contents of which is incorporated herein by reference.

### **BACKGROUND**

- (1) The present disclosure relates generally to medical fluid connectors and, more particularly, to self-sealing fluid connector systems having first and second connectors that can be coupled together to form a fluid pathway and to obstruct the fluid pathway upon separation of the first and second connectors.
- (2) Medical connections are widely used in fluid delivery systems such as those used in connection with intravenous (IV) fluid lines, blood access, hemodialysis, peritoneal dialysis, enteral feeding, drug vial access, and other procedures.
- (3) In some instances, the medical connection can become dislodged or disconnected in an unintended manner. For example, medical tubing of an IV set that is coupled to a catheter can

become dislodged when an unintended or unexpected forces is exerted upon the catheter, which may exceed the design limitations of the catheter securement method. An unintended or unexpected force can be applied to the tubing and/or catheter when the patient moves or rolls over within a bed, or when the tubing or another portion of an intravenous set become caught on a portion of the bed, such as the railing, or when a patient is panicking, disoriented, or fidgeting to such an extent that the medical tubing is unintentionally or intentionally pulled away from the patient or away from the medical equipment coupled to the tubing.

## SUMMARY

(4) In accordance with at least some embodiments disclosed herein is the realization that unintended dislodgement or disconnection of a medical connection, such as a medical fluid line, can result in injury to a patient or a caregiver, such as by depriving the patient of a medicament, increasing the potential for infection to the patient, and exposing the caregiver to harmful medicaments.

(5) Accordingly, aspects of the present disclosure provide fluid connector system comprising a first connector and a second connector, the first connector comprising a housing having a proximal end and a distal end, an inner surface forming a housing channel extending between a proximal opening and a distal opening of the housing, and a first valve positioned in the housing channel, and the second connector comprising a body having a proximal end and a distal end, an inner surface forming a body channel extending between a proximal opening and a distal opening of the body, and a second valve positioned in the body channel, wherein an outer surface of the distal end of the body forms a protrusion and one or more arm, wherein the one or more arm extends along the protrusion, and wherein, when the first and second connectors are coupled together, the one or more arm is engaged against the housing to resist separation of the first and second connectors, and the protrusion extends to the proximal opening of the housing to engage against the first valve and open a fluid passageway through the first and second connectors.

(6) In some instances, the present disclosure includes a method for providing a fluid connector system, the method comprising providing a first connector comprising a housing having a proximal end and a distal end, an inner surface forming a housing channel extending between a proximal opening and a distal opening of the housing, and a first valve positioned in the channel of the housing, and providing a second connector comprising a body having a proximal end and a distal end, an inner surface forming a body channel extending between a proximal opening and a distal opening of the body, and a second valve positioned in the channel of the body, wherein an outer surface of the distal end of the body forms a protrusion and one or more arm, wherein the one or more arm extends along the protrusion, wherein the first and second connectors are couplable together such that the one or more arm engages against the housing and the protrusion extends to the proximal opening of the housing to engage against the first valve and open a fluid passageway through the first and second connectors.

(7) Accordingly, the present application addresses several operational challenges encountered in prior fluid connections and provides numerous improvements that enable the user to increase safety and efficacy, while more easily and precisely providing fluid connections.

(8) Additional features and advantages of the subject technology will be set forth in the description below, and in part will be apparent from the description, or may be learned by practice of the subject technology. The advantages of the subject technology will be realized and attained by the structure particularly pointed out in the written description and embodiments hereof as well as the appended drawings.

(9) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are intended to provide further explanation of the subject technology.

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# Description

## BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various features of illustrative embodiments of the inventions are described below with reference to the drawings. The illustrated embodiments are intended to illustrate, but not to limit, the inventions. The drawings contain the following figures:
- (2) FIG. 1 illustrates a fluid connector system in use with an IV set coupled to a patient, in accordance with aspects of the present disclosure.
- (3) FIG. 2 illustrates a perspective view of an embodiment of a fluid connector system, in accordance with aspects of the present disclosure.
- (4) FIG. 3 illustrates a cross-sectional view of the fluid connector system of FIG. 2, in accordance with aspects of the present disclosure.
- (5) FIG. 4 illustrates a cross-sectional view of the fluid connector system of FIG. 2, in accordance with aspects of the present disclosure.
- (6) FIG. 5 illustrates a detail view of the fluid connector system of FIG. 3, in accordance with aspects of the present disclosure.
- (7) FIG. 6 illustrates a cross-sectional view of another embodiment of a fluid connector system, in accordance with aspects of the present disclosure.
- (8) FIG. 7 illustrates an exploded cross-sectional view of a first connector of the fluid connector system, in accordance with aspects of the present disclosure.
- (9) FIG. 8 illustrates an exploded cross-sectional view of a second connector of the fluid connector system, in accordance with aspects of the present disclosure.

## DETAILED DESCRIPTION

- (10) In the following detailed description, numerous specific details are set forth to provide a full understanding of the subject technology. It should be understood that the subject technology may be practiced without some of these specific details. In other instances, well-known structures and techniques have not been shown in detail so as not to obscure the subject technology.
- (11) Further, while the present description sets forth specific details of various embodiments, it will be appreciated that the description is illustrative only and should not be construed in any way as limiting. Additionally, it is contemplated that although particular embodiments of the present disclosure may be disclosed or shown in the context of an IV set, such embodiments can be used in other fluid conveyance systems. Furthermore, various applications of such embodiments and modifications thereto, which may occur to those who are skilled in the art, are also encompassed by the general concepts described herein.
- (12) In accordance with some embodiments, the present application discloses various features and advantages of a fluid connector system. The fluid connector system can provide for efficient and safe maintenance of fluid connections, such as the connections used for transferring medical fluids toward or away from a patient. The fluid connector system can maintain a fluid pathway by resisting unintended disconnection when a pulling or tension force is applied to the fluid connector system, such as when a patient moves or when the medical tubing is pulled away from the patient.
- (13) The fluid connector system can also prevent injury to a patient or a caregiver by permitting disconnection or separation between portions of the connector system when a pulling or tension force exceeds a threshold. The fluid connector system can also prevent injury to a patient or a caregiver by obstructing the fluid pathway when disconnection or separation between portions of the connector system occurs. Further, the fluid connector system can provide for efficient and safe reestablishment of the fluid pathway, by permitting reassembly of portions of the system after a disconnection or separation occurs.
- (14) Referring now to the figures, FIG. 1 illustrates an example of a fluid connector system in use in accordance with aspects of the present disclosure. The fluid connector system **100** is coupled

with tubing of an IV set, which is being used to direct a fluid to a patient **10**. The IV set can include a first length of tubing **16A** that is Coupled to a first portion of the fluid connector system **100**, and a second length of tubing **16B** that is coupled to fluid connector system **100** and extends to an IV catheter **18**.

(15) The fluid connector system **100** couples the first and second lengths of tubing **16A**, **16B** to the IV catheter **18** so that a fluid can move through the tubing **16A**, **16B** and the IV catheter **18** by flowing through the fluid connector system **100**. If a sufficient pulling or tension force is applied to the fluid connector system **100**, such as when a patient moves or when the medical tubing is pulled away from the patient, portions of the fluid connector system **100** can separate from each other to separate the first length of tubing **16A** from the patient and resist fluid flow out of the first length of tubing **16A** and out of the second length of tubing **16B**.

(16) Although the fluid connector system **100** is illustrated being coupled along a fluid pathway of an IV set, it should be understood that the fluid connector system **100** can be connected within other fluid pathways, such as between a patient and a IV pump or between a patient and a dialysis machine. The fluid connector system **100** can also be connected along another portion of a fluid pathway. For example, the fluid connector system **100** can be connected along a proximal portion of the fluid pathway, such as being connected between the tubing **16** and the medicament bag **12** or other fluid therapy device. In another example, any of the first and second portions of the fluid connector system **100** can be directly coupled to another fluid delivery devices, such as the catheter or a medicament bag.

(17) The portions of the fluid connector system **100** that are separable from each other include a first connector **110** and a second connector **210**, which are illustrated in FIGS. **1** and **2**. The first and second connectors **110**, **210** can be coupled together by engaging a portion of the second connector **210** against a portion of the first connector **110**. When the first and second connectors **110**, **210** coupled together, a fluid channel of each of the first and second connectors **110**, **210** is coupled together to form a fluid pathway through the fluid connector system **100**.

(18) The first and second connectors **110**, **210** resist unintended separation from each other. However, if any of the first connector **110** and the second connector **210** are separated from each other, such as when a pulling or tension force on the fluid connector system **100** exceeds a threshold, the first connector **110** and the second connector **210** can separate from each other while maintaining the ability to reconnect the first and second connectors **110**, **210** together. In some embodiments of the present disclosure, the threshold force for separating the first and second connectors **110**, **210** is greater than or equal to approximately five pounds (22.25 Newtons).

(19) When the first and second connectors **110**, **210**, separate from each other, as illustrated in FIG. **3**, the respective first and second connectors **110**, **210** can resist a fluid flow therethrough. In some embodiments of the present disclosure, each of the first and second connectors **110**, **210** can resist fluid flow by obstructing a fluid passage through their respective assemblies.

(20) The first connector **110** includes a housing **112** having a proximal end **114**, a distal end **116**, and an inner surface **120** forming a housing channel **122** extending between a proximal opening **124** at the proximal end **114** of the housing and a distal opening **126** at the distal end **116** of the housing. A proximal end of the housing **112**, forming the proximal end **114** and the proximal opening **124**, is configured to removable couple with the second connector **210**.

(21) A first valve **140** is positioned in the housing channel **122** to resist or permit a fluid flow through the housing channel **122**. When the first connector **110** is not coupled with the second connector **210** or another mating connector, as shown in FIG. **4**, the first valve **140** is in a closed position. In the closed position, the housing channel **122** is obstructed by the first valve **140** to resist fluid flow through the first connector **110**.

(22) When the first connector **110** is coupled with the second connector **210**, as shown in at least FIGS. **2** and **3**, the first valve **140** is moved to an open position. In the open position, the first valve **140** is configured to impart no resistance, or less resistance relative to the opening position of the

first valve **140**. Thus, in the open position of the first valve **140**, the housing channel **122** is unobstructed to permit a fluid flow through the first connector **110**.

(23) The first valve **140** of the first connector can be configured as a check valve that can permit a fluid to move through the first valve **140** in an open position, and can resist movement of the fluid through the first valve **140** in an closed position. In the open position, the first valve **140** does not resist movement of a fluid in a direction from the proximal end **114** toward the distal end **116** of the housing, and in the closed position, the first valve **140** resists movement of a fluid in a direction from the distal end **116** toward the proximal end **114** of the housing.

(24) The first valve **140** is shaped, in some embodiments, as a duck-billed check valve; however, the present disclosure contemplates that the first valve **140** can be configured as check valve having another shape or structure.

(25) The second connector **210** includes a body **212** having a proximal end **214**, a distal end **216**, and an inner surface **220** forming a body channel **222** extending between a proximal opening **224** at the proximal end **214** of the housing and a distal opening **226** at the distal end **216** of the body.

(26) A portion of the body **212**, including the distal end **216** and the distal opening **226**, forms a protrusion **240** and one or more arm **260**, where the protrusion **240** is configured to extend toward the proximal opening **124** of the housing and the one or more arm **260** is configured to engage against the housing **112** when the first and second connectors **110**, **210** are coupled together.

(27) The protrusion **240** is also configured to form a portion of the fluid passageway through the second connector **210**, and therefore, the inner surface **220** of the body **212** forms a portion of the body channel **222** through the protrusion **240**. The body channel **222** extends to the distal opening **226** of the body, which is located at a distal end portion of the protrusion **240**. In some embodiments of the present disclosure, the distal opening **226** is located at a distal-most end of the protrusion **240**; however, it should be understood that the present disclosure contemplated embodiments in which the distal opening **226** is positioned along another portion of the protrusion **240**. For example, in some embodiments of the present disclosure, the distal opening **226** is located between a proximal and distal ends of the protrusion **240**.

(28) The one or more arm **260** extends in a direction away from the proximal end **214** of the body such that, when the first and second connectors **110**, **210** are coupled together, a portion of the one or more arm **260** engages against the housing **112** of the first connector to resist movement of the first and second connectors **110**, **210** in a direction away from each other. Engagement of the one or more arm **260** against the housing **112** can, in some instances of the present disclosure, define a snap fitting or snap joint between the first and second connectors **110**, **210**.

(29) The one or more arm **260** includes an inner surface **262** that is spaced apart from the outer surface **242** of the protrusion by a distance to permit a portion of the first connector **110** to be positioned between the protrusion **240** and the one or more arm **260** when the first and second connectors **110**, **210** are coupled together.

(30) To resist movement of the first and second connectors **110**, **210** in a direction away from each other, the one or more arm **260** can include a ridge **264** at a distal end portion **266** of the one or more arm. To resist movement of the first and second connectors **110**, **210**, the ridge **264** is configured to engage against the first connector **110**. To engage against the first connector **110**, the ridge **264** extends radially inward in a direction toward the protrusion **240** of the second connector.

(31) Although the ridge **264** is configured to resist separation of the first and second connectors **110**, **210**, the one or more arm **260** is also configured to permit separation of the first and second connectors **110**, **210** when a threshold force exceeded between the first and second connectors **110**, **210**. Separation of the first and second connectors **110**, **210** can occur when the one or more arm **260** is biased or flexed in a direction away from the protrusion **240**.

(32) the one or more arm **260**, or the distal end portion **266** of the one or more arm, is caused to bias or flex by engagement of the ridge **264** against the housing **112** when the first and second connectors **110**, **210** are moved away from each other.

(33) In some embodiments of the present disclosure, the one or more arm **260** comprises a first arm **261A** and a second arm **261B**. As illustrated, for example, FIGS. **3** and **4** the first and second arms **261A**, **261B** can be spaced apart from each other around a longitudinal axis **A2** extending between the proximal and distal ends of the body.

(34) In some embodiments of the present disclosure, the one or more arm **260** is shaped as a continuous or semi-continuous circumferential wall **270** extending around a longitudinal axis **A2**, as illustrated, for example, FIGS. **2** and **6**. In some instances of the present disclosure, the first and second arms **261A**, **261B** can be defined by the circumferential wall **270** having one or more channel extending therethrough.

(35) A second valve **280** is positioned in the body channel **222** to resist a fluid flow through the body channel **222** in the closed position, and permit fluid flow through the body channel **222** in the open position. In some embodiments, of the present disclosure, the second valve **280** is in the closed position when the second connector **210** is not coupled with the first connector **110**, and is moved to the open position when the first and second connectors are coupled together. In some embodiments, the second valve **280** is moved from the closed to the open position by pressure of a fluid in the body channel **222**.

(36) The second valve **280** can be configured as a check valve that can permit a fluid to move through the second valve **280** in an open position, and can resist movement of the fluid through the second valve **280** in a closed position. In the open position, the second valve **280** does not resist movement of a fluid between the proximal end **214** and the distal end **216** of the body, and in the closed position, the second valve **280** resists movement of a fluid between the proximal end **214** and the distal end **216** of the body.

(37) The second valve **280** is shaped, in some embodiments, as a duck-billed check valve; however, the present disclosure contemplates that the second valve **280** can be configured as check valve having another shape or structure.

(38) In some embodiments of the present disclosure, any of the distal opening **126** of the first connector and the proximal opening **224** of the second connector can be configured as any of a male and female luer.

(39) In some embodiments, a distal end portion of the housing **112** that forms the distal opening **126** can have an outer surface shaped as a male luer, and a proximal end portion of the body **212** that forms the proximal opening **224** can be shaped as a female luer. Any of the male and female luer can be configured to couple with any of a tubing or mating connector.

(40) To couple with a tubing or mating connector, the first connector **110** and/or second connector **210** forming the male luer also includes a sleeve **128** having an inner surface that is spaced apart from the outer surface of the male luer. The sleeve **128** is configured to engage against a mating connector coupled to the male luer. In some instances of the present disclosure, the sleeve **128** includes a thread **130** extending along an inner surface of the sleeve and configured to mate with a thread of the mating connector. In some instances, the present disclosure also contemplates that the sleeve **128** can be configured to couple with a mating connector using an interference fit therebetween, or using another mating structure, such as a notch and groove.

(41) In some embodiments of the present disclosure, the first connector **110** can also include an insert **160** coupled to the proximal opening **124** and configured to retain the first valve **140** with the housing **112**. The features of **110**, the insert **160** can include a first end **162**, a second end **164**, an inner surface **166** forming a cavity **168** extending into the second end **164**, and a passage **170** extending through the first end **162** and intersecting the cavity **168**.

(42) The first valve **140** is positioned between the insert **160** and the housing **112**, and is oriented adjacent to the passage **170**. In some embodiments of the present disclosure, a first portion of the first valve **140** is positioned within the cavity **168** of the insert, and a second portion of the first valve **140** is positioned within the housing channel **122**. Additionally, a portion of the first valve **140** can intersect a passageway defined between the passage **170** and the housing channel **122**.

(43) When coupled to the proximal opening **124** of the housing, the first end **162** of the insert can receive a portion of the second connector **210** through the passage **170** to form a portion of the fluid passageway through the fluid connector system **100**. To prevent leakage or a contaminant from moving between the first and second connectors **110**, **210**, and into the fluid passageway through the fluid connector system **100**, a seal **172** is positioned along the passage **170** of the insert. It should be understood, however, that although a seal **172** is disclosed herein, the present disclosure contemplates that a seal between the first and second connectors **110**, **210**, or between features of each of the first and second connectors **110**, **210**, by an interference fit, a sealant substance, or an adhesive.

(44) A detail view of the interface between the one or more arm **260** of the body **212** of the second connector and the housing **112** of the first connector is illustrated in the detail view of FIG. 5. The interface between the one or more arm **260** and the housing **112** can be defined by engagement of the ridge **264** of the one or more arm against a flange **132** of housing **112**.

(45) The flange **132** can be located along a portion of the housing **112** forming the proximal end **114** and the proximal opening **124**. When the first and second connectors **110**, **210** are coupled together, the ridge **264** is positioned between the flange **132** and the distal end **116** of the housing.

(46) Although in some embodiments of the present disclosure, the flange **132** is configured to extend radially outward in a direction away from a longitudinal axis **A1** extending between the proximal and distal ends of the housing, the present disclosure contemplates other structures to resist movement of the one or more arm **260**. In some embodiments of the present disclosure, the first connector **110** can include a structure to resist movement of the one or more arm **260** of the second connector, including, for example, a channel that extends into the outer surface of the housing **112** and is configured to receive the ridge **264** therein.

(47) FIG. 5 also illustrates embodiments of the present disclosure in which the flange **132** comprises a distal-facing surface **134** and a proximal-facing surface **136**, and the ridge **264** comprises a distal facing surface **256** and a proximal-facing surface **258**.

(48) When moved in a direction toward each other, the distal facing surface **256** of the ridge can engage against the proximal-facing surface **136** of the flange. Engagement of the distal facing surface **256** of the ridge against the proximal-facing surface **136** of the flange can cause the one or more arm **260** to bias or flex radially outward, thereby permitting the first and second connectors **110**, **210** to be moved further toward each other.

(49) When the first and second connectors **110**, **210** are moved toward each other by a distance such that the ridge is distal to the flange **132**, the one or more arm **260** can move radially inward to a less-biased or less-flexed position.

(50) When the first and second connectors **110**, **210** are coupled together, the distal-facing surface **134** of the flange can be aligned with the proximal-facing surface **258** of the ridge such that a common plane **P1** extends along each of the distal-facing surface **134** and the proximal-facing surface **258**.

(51) The common plane **P1** can be configured by forming the distal-facing surface **134** of the flange at a first angle, relative to the longitudinal axis **A1**, and the proximal-facing surface **258** of the ridge at a second angle, relative to the longitudinal axis **A2**. In some aspects of the present disclosure, the first and second angles are approximately 40 degrees relative to the longitudinal axis **A1** of the housing **112**. The first and second angles can, in some instances, be selected to increase or decrease resistance to assembly and separation between the first and second connectors **110**, **210**.

(52) The resistance or force to assembly and separation between the first and second connectors **110**, **210**, as a result of the first and second angles, can be configured so that the force required for assembly of the first and second connectors **110**, **210** is less than the force required for separation of the first and second connectors **110**, **210**. In some embodiments of the present disclosure, the force required to cause deflection or biasing of the one or more arm such that the first and second connectors **110**, **210** are separable is approximately two times greater than the force required to cause

deflection or biasing of the one or more arm such that the first and second connectors **110**, **210** can be coupled together.

(53) Although in some embodiments of the present disclosure, the first and second angles can provide a common plane **P1** between the distal-facing surface **134** and the proximal-facing surface **258**, the present disclosure also contemplates embodiments in which each of the distal-facing surface **134** and the proximal-facing surface **258** define a plane that does not intersect.

(54) Assembly of embodiments of the first and second connectors **110**, **210** is illustrated in Figures **7** and **8**, respectively. Referring to FIG. **7**, the first connector **110** can be assembled by inserting the first valve **140** into the proximal opening **124** of the housing **112**. The insert **160** can be moved toward the proximal end **114** of the housing with the first valve **140** therebetween. In some embodiments of the present disclosure, the first valve **140** is retained in the first connector **110** when the second end **164** of the insert is coupled to the proximal end **114** of the housing **112**. The seal **172** of the first connector can be inserted into the passage **170**, and in some embodiments, the insert **160** can include a groove configured to receive the seal **172** therein.

(55) Referring to FIG. **8**, the second connector **210** can be assembled by inserting the second valve **280** into the body channel **222** of the body. The second valve **280** can be inserted into the body channel **222** through any of the proximal opening **224** at the proximal end of the body or through another opening of the body. In some embodiments, however, the body **212** comprises a distal body portion **213B** and a proximal body portion **213A**, where the distal and proximal body portions **213B**, **213A** can be coupled together to form the body channel **222** therebetween. Additionally, the distal and proximal body portions **213B**, **213A** can be coupled together with the second valve **280** positioned along the body channel **222** therebetween.

(56) The features of the present disclosure provide first and second connectors that can be coupled together to form a fluid pathway therebetween. When coupled together, the features of the present disclosure resist unintentional separation between the first and connectors. However, if the first and connectors are separated, whether unintentionally or intentionally, the fluid pathway for each of the first and connectors become closed or obstructed to prevent fluid loss therefrom. The features of the present disclosure as **1** provide that upon separation of the first and second connectors, any of the first and connectors can be cleaned and disinfected, and the first and second connectors can be once again coupled together to form a fluid pathway therebetween.

(57) Illustration of Subject Technology as Clauses

(58) The subject technology is illustrated, for example, according to various aspects described below. Various examples of aspects of the subject technology are described as numbered clauses (1, 2, 3, etc.) for convenience. These are provided as examples and do not limit the subject technology. It is noted that any of the dependent clauses may be combined in any combination, and placed into a respective independent clause, e.g., clause 1 or clause 5. The other clauses can be presented in a similar manner.

(59) Clause 1. A fluid connector system comprising: a first connector comprising: a housing having a proximal end and a distal end, an inner surface forming a housing channel extending between a proximal opening and a distal opening of the housing, and a first valve positioned in the housing channel; and a second connector comprising: a body having a proximal end and a distal end, an inner surface forming a body channel extending between a proximal opening and a distal opening of the body, and a second valve positioned in the body channel, wherein an outer surface of the distal end of the body forms a protrusion and one or more arm, wherein the one or more arm extends along the protrusion; wherein, when the first and second connectors are coupled together, the one or more arm is engaged against the housing to resist separation of the first and second connectors, and the protrusion extends to the proximal opening of the housing to engage against the first valve and open a fluid passageway through the first and second connectors.

(60) Clause 2. The fluid connector system of Clause 1, wherein the first connector comprises an insert comprising a first end, a second end, an inner surface forming a cavity extending into the

second end, and a passage extending through the first end and intersecting the cavity, and wherein the insert is coupled to the proximal end of the housing with the first valve positioned between the cavity of the insert and the proximal opening of the housing.

(61) Clause 3. The fluid connector system of Clause 2, wherein, when the first and second connectors are coupled together the protrusion extends through the passage of the insert.

(62) Clause 4. The fluid connector system of Clause 2, wherein the insert comprises a seal positioned along the passage and configured to engage against the protrusion when the first and second connectors are coupled together.

(63) Clause 5. The fluid connector system of Clause 2, wherein the second end of the insert extends into the proximal opening of the housing.

(64) Clause 6. The fluid connector system of any one of Clauses 1 to 5, wherein the first valve is a check valve configured to permit a fluid to move through the check valve in a direction from the proximal end toward the distal end of the housing, to resist movement of the fluid through the check valve in a direction from the distal end toward the proximal end of the housing.

(65) Clause 7. The fluid connector system of any one of Clauses 1 to 6, wherein an outer surface of the distal end of the housing forms a male luer shape with the housing channel extending therethrough.

(66) Clause 8. The fluid connector system of Clause 7, wherein the distal end of the housing comprises a sleeve having an inner surface that is spaced apart from the outer surface of the male luer.

(67) Clause 9. The fluid connector system of Clause 8, wherein the housing comprises a thread extending along the inner surface of the sleeve.

(68) Clause 10. The fluid connector system of any one of Clauses 1 to 9, wherein an outer surface of the housing comprises a flange that extends radially outward in a direction away from a longitudinal axis extending between the proximal and distal ends of the housing.

(69) Clause 11. The fluid connector system of Clause 10, wherein, when the first and second connectors are coupled together, a portion of the one or more arm is positioned between the flange and the distal end of the housing.

(70) Clause 12. The fluid connector system of Clause 10, wherein the flange comprises a distal-facing surface, the one or more arm comprises a ridge having a proximal facing surface, and wherein, when the first and second connectors are coupled together, the distal-facing surface and the proximal facing surface are coplanar.

(71) Clause 13. The fluid connector system of any one of Clauses 1 to 12, the one or more arm comprises a ridge that extends radially inward in a direction toward a longitudinal axis extending between the proximal and distal ends of the body.

(72) Clause 14. The fluid connector system of any one of Clauses 1 to 13, wherein the second valve is a check valve configured to permit a fluid to move through the check valve in a direction from the proximal end toward the distal end of the body, to resist movement of the fluid through the check valve in a direction from the distal end toward the proximal end of the body.

(73) Clause 15. The fluid connector system of any one of Clauses 1 to 14, wherein the one or more arm comprises a first arm and a second arm, and wherein the first and second arm are spaced apart around a longitudinal axis extending between the proximal and distal ends of the body.

(74) Clause 16. The fluid connector system of any one of Clauses 1 to 15, wherein the one or more arm comprises a circumferential wall having an inner surface that is spaced apart from the outer surface of the protrusion.

(75) Clause 17. A method for providing a fluid connector system, the method comprising: providing a first connector comprising a housing having a proximal end and a distal end, an inner surface forming a housing channel extending between a proximal opening and a distal opening of the housing, and a first valve positioned in the channel of the housing; and providing a second connector comprising a body having a proximal end and a distal end, an inner surface forming a

body channel extending between a proximal opening and a distal opening of the body, and a second valve positioned in the channel of the body, wherein an outer surface of the distal end of the body forms a protrusion and one or more arm, wherein the one or more arm extends along the protrusion; wherein the first and second connectors are couplable together such that the one or more arm engages against the housing and the protrusion extends to the proximal opening of the housing to engage against the first valve and open a fluid passageway through the first and second connectors.

(76) Clause 18. The method of Clause 17, wherein the body comprises a proximal portion and a distal portion, and providing the second valve comprises coupling the proximal and distal portions of the body together with the second valve therebetween.

(77) Clause 19. The method of Clause 17, further comprising providing an insert configured to couple to the proximal end of the housing.

(78) Clause 20. The method of Clause 19, wherein providing the first valve comprises coupling the insert to the proximal end of the housing with the first valve positioned therebetween.

#### (79) Further Considerations

(80) In some embodiments, any of the clauses herein may depend from any one of the independent clauses or any one of the dependent clauses. In one aspect, any of the clauses (e.g., dependent or independent clauses) may be combined with any other one or more clauses (e.g., dependent or independent clauses). In one aspect, a claim may include some or all of the words (e.g., steps, operations, means or components) recited in a clause, a sentence, a phrase or a paragraph. In one aspect, a claim may include some or all of the words recited in one or more clauses, sentences, phrases or paragraphs. In one aspect, some of the words in each of the clauses, sentences, phrases or paragraphs may be removed. In one aspect, additional words or elements may be added to a clause, a sentence, a phrase or a paragraph. In one aspect, the subject technology may be implemented without utilizing some of the components, elements, functions or operations described herein. In one aspect, the subject technology may be implemented utilizing additional components, elements, functions or operations.

(81) The present disclosure is provided to enable any person skilled in the art to practice the various aspects described herein. The disclosure provides various examples of the subject technology, and the subject technology is not limited to these examples. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects.

(82) A reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. Pronouns in the masculine (e.g., his) include the feminine and neuter gender (e.g., her and its) and vice versa. Headings and subheadings, if any, are used for convenience only and do not limit the invention.

(83) The word “exemplary” is used herein to mean “serving as an example or illustration.” Any aspect or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs. In one aspect, various alternative configurations and operations described herein may be considered to be at least equivalent.

(84) A phrase such as an “aspect” does not imply that such aspect is essential to the subject technology or that such aspect applies to all configurations of the subject technology. A disclosure relating to an aspect may apply to all configurations, or one or more configurations. An aspect may provide one or more examples. A phrase such as an aspect may refer to one or more aspects and vice versa. A phrase such as an “embodiment” does not imply that such embodiment is essential to the subject technology or that such embodiment applies to all configurations of the subject technology. A disclosure relating to an embodiment may apply to all embodiments, or one or more embodiments. An embodiment may provide one or more examples. A phrase such an embodiment may refer to one or more embodiments and vice versa. A phrase such as a “configuration” does not imply that such configuration is essential to the subject technology or that such configuration

applies to all configurations of the subject technology. A disclosure relating to a configuration may apply to all configurations, or one or more configurations. A configuration may provide one or more examples. A phrase such a configuration may refer to one or more configurations and vice versa.

(85) In one aspect, unless otherwise stated, all measurements, values, ratings, positions, magnitudes, sizes, and other specifications that are set forth in this specification, including in the claims that follow, are approximate, not exact. In one aspect, they are intended to have a reasonable range that is consistent with the functions to which they relate and with what is customary in the art to which they pertain.

(86) In one aspect, the term “coupled” or the like may refer to being directly coupled. In another aspect, the term “coupled” or the like may refer to being indirectly coupled.

(87) Terms such as “top,” “bottom,” “front,” “rear” and the like if used in this disclosure should be understood as referring to an arbitrary frame of reference, rather than to the ordinary gravitational frame of reference. Thus, a top surface, a bottom surface, a front surface, and a rear surface may extend upwardly, downwardly, diagonally, or horizontally in a gravitational frame of reference.

(88) Various items may be arranged differently (e.g., arranged in a different order, or partitioned in a different way) all without departing from the scope of the subject technology. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112, sixth paragraph, unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.” Furthermore, to the extent that the term “include,” “have,” or the like is used, such term is intended to be inclusive in a manner similar to the term “comprise” as “comprise” is interpreted when employed as a transitional word in a claim.

(89) The Title, Background, Summary, Brief Description of the Drawings and Abstract of the disclosure are hereby incorporated into the disclosure and are provided as illustrative examples of the disclosure, not as restrictive descriptions. It is submitted with the understanding that they will not be used to limit the scope or meaning of the claims. In addition, in the Detailed Description, it can be seen that the description provides illustrative examples and the various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed subject matter requires more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed configuration or operation. The following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

(90) The claims are not intended to be limited to the aspects described herein, but is to be accorded the full scope consistent with the language claims and to encompass all legal equivalents. Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirement of 35 U.S.C. § 101, 102, or 103, nor should they be interpreted in such a way.

## Claims

1. A fluid connector system comprising: a first connector comprising: a housing having a proximal end and a distal end, an inner surface forming a housing channel extending between a proximal opening and a distal opening of the housing, an insert coupled with the proximal opening, and a first valve positioned in the housing channel and between the insert and the housing; and a second connector comprising: a body having a proximal end and a distal end, an inner surface forming a

body channel extending between a proximal opening and a distal opening of the body, and a second valve positioned in the body channel, wherein an outer surface of the distal end of the body forms a protrusion and one or more arm, wherein the one or more arm extends along the protrusion; wherein, when the first and second connectors are coupled together, the one or more arm is engaged against the housing to resist separation of the first and second connectors, and the protrusion extends to the proximal opening of the housing to engage against the first valve and open a fluid passageway through the first and second connectors.

2. The fluid connector system of claim 1, wherein the insert comprises a first end, a second end, an inner surface forming a cavity extending into the second end, and a passage extending through the first end and intersecting the cavity, and wherein the insert is coupled to the proximal end of the housing with the first valve positioned between the cavity of the insert and the proximal opening of the housing.

3. The fluid connector system of claim 2, wherein, when the first and second connectors are coupled together the protrusion extends through the passage of the insert.

4. The fluid connector system of claim 2, wherein the insert comprises a seal positioned along the passage and configured to engage against the protrusion when the first and second connectors are coupled together.

5. The fluid connector system of claim 2, wherein the second end of the insert extends into the proximal opening of the housing.

6. The fluid connector system of claim 1, wherein the first valve is a check valve configured to permit a fluid to move through the check valve in a direction from the proximal end toward the distal end of the housing, to resist movement of the fluid through the check valve in a direction from the distal end toward the proximal end of the housing.

7. The fluid connector system of claim 1, wherein an outer surface of the distal end of the housing forms a male luer shape with the housing channel extending therethrough.

8. The fluid connector system of claim 7, wherein the distal end of the housing comprises a sleeve having an inner surface that is spaced apart from the outer surface of the male luer.

9. The fluid connector system of claim 8, wherein the housing comprises a thread extending along the inner surface of the sleeve.

10. The fluid connector system of claim 1, wherein an outer surface of the housing comprises a flange that extends radially outward in a direction away from a longitudinal axis extending between the proximal and distal ends of the housing.

11. The fluid connector system of claim 10, wherein, when the first and second connectors are coupled together, a portion of the one or more arm is positioned between the flange and the distal end of the housing.

12. The fluid connector system of claim 10, wherein the flange comprises a distal-facing surface, the one or more arm comprises a ridge having a proximal facing surface, and wherein, when the first and second connectors are coupled together, the distal-facing surface and the proximal facing surface are coplanar.

13. The fluid connector system of claim 1, wherein the one or more arm comprises a ridge that extends radially inward in a direction toward a longitudinal axis extending between the proximal and distal ends of the body.

14. The fluid connector system of claim 1, wherein the second valve is a check valve configured to permit a fluid to move through the check valve in a direction from the proximal end toward the distal end of the body, to resist movement of the fluid through the check valve in a direction from the distal end toward the proximal end of the body.

15. The fluid connector system of claim 1, wherein the one or more arm comprises a first arm and a second arm, and wherein the first and second arm are spaced apart around a longitudinal axis extending between the proximal and distal ends of the body.

16. The fluid connector system of claim 1, wherein the one or more arm comprises a

circumferential wall having an inner surface that is spaced apart from the outer surface of the protrusion.

17. A method for providing a fluid connector system, the method comprising: providing a first connector comprising a housing having a proximal end and a distal end, an inner surface forming a housing channel extending between a proximal opening and a distal opening of the housing, an insert coupled with the proximal opening, and a first valve positioned in the channel of the housing and between the insert and the housing; and providing a second connector comprising a body having a proximal end and a distal end, an inner surface forming a body channel extending between a proximal opening and a distal opening of the body, and a second valve positioned in the channel of the body, wherein an outer surface of the distal end of the body forms a protrusion and one or more arm, wherein the one or more arm extends along the protrusion; wherein the first and second connectors are couplable together such that the one or more arm engages against the housing and the protrusion extends to the proximal opening of the housing to engage against the first valve and open a fluid passageway through the first and second connectors.

18. The method of claim 17, wherein the body comprises a proximal portion and a distal portion, and providing the second valve comprises coupling the proximal and distal portions of the body together with the second valve therebetween.

19. The method of claim 17, wherein providing the first valve comprises coupling the insert to the proximal end of the housing with the first valve positioned therebetween.

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